Fiscal Year:	FY 2015	**Task Last Updated:**	FY 08/29/2014
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Genes that Predict the Loss of Bone during Weightlessness		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	stefan.judex@sunysb.edu	**Fax:**	FY 631-632-8577
PI Organization Type:	UNIVERSITY	**Phone:**	631-632-1549
Organization Name:	SUNY Stony Brook		
PI Address 1:	Dept. of Biomedical Engineering		
PI Address 2:	Bioengineering Building 213		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	11/01/2012	**End Date:**	10/31/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Leah Rae (Jackson Laboratory)		
Grant/Contract No.:	NNX12AL25G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The amount of bone loss between individual astronauts is highly variable with some astronauts losing large amounts of bone while others are largely unaffected. This variability is accounted for, at least to a large extent, by genetic variations between individuals. Unfortunately, the specific genes that modulate this genetic sensitivity of bone loss in microgravity are unknown. To this end, we have recently completed a genetic study in which we identified specific regions on chromosomes (QTL or Quantitative Trait Loci) that either add to or reduce the propensity of an individual to lose bone when normal weightbearing is removed from the musculoskeleton. While providing critical information, these QTL still contain hundreds of genes, rather than a single one. Here, we propose first to greatly reduce the number of candidate genes with bioinformatics tools and then experimentally test the shortlist of candidate genes using knockout mice available from the Jackson Laboratory's (JAX) Genetic Resource Repository and the JAX Knock Out Mouse Project (KOMP) production center. The identification of single genes modulating bone loss during weightlessness can be used		

	as a critical diagnostic tool for the identification of astronauts most in need of countermeasures as well as to unravel novel drug targets that preserve bone health in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal directly addresses one of the six highest priority recommendations made in the National Research Council's 2011 Decadal Survey Report, "Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era" (http://www.nap.edu/), namely recommendation AH3: "Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended." Defining some of the specific genes that cause differences in the skeletal sensitivity to weightlessness will not only provide novel diagnostic and therapeutic means of combating bone loss in space but also on Earth in individuals subjected to bedrest or suffering from paralysis.
Task Progress:	Progress of Specific Aim 1: Specific Aim 1 was completed in the first year of this grant. Progress of Specific Aim 2 Introduction Specific Aim 1 provided a short of list of candidate genes that may modulate the magnitude of bone's response to unloading and reambulation. Over the past 12 months, we continued to test specific transgenic mouse models for their degree of bone's susceptibility to unloading. Recent targets were fibroblast growth factor 2 (FGF-2), estrogen receptor 1 (ESR-1), and nucleotide binding oligomerization domain containing protein 2 (NOD-2). Previous in vitro studies indicated a role for FGF2 in proliferation and differentiation of cells (1). Specifically, FGF2 is expressed in osteoblasts and alteration in expression levels of FGF2 leads to an imbalance in bone remodeling. For instance, overexpression of FGF2 stunted growth and induced chondrodysplasia (2, 3). Further, administration of fibroblast growth factor can be anabolic in ovariectomized rats while parathyroid hormone administration lost its efficacy in the absence of FGF2 (4, 5). We therefore hypothesized that impaired bone remodeling following loss of FGF2 function will be further exacerbated during disuse. Another one of our candidate genes, estrogen receptor 1 (ESR1) is the primary receptor for estrogen. Bone loss after menopause or surgical removal of ovaries has been well established (6). Moreover, bone's anabolic response to mechanical stimuli is reduced when estrogen signaling though ESR1 is blocked (7). We hypothesized that bone's diminished response to mechanical signals would lead to an impaired response to the loss of mechanical signals, resulting in less bone loss during unloading (8). NOD2 is an intracellular protein associated with inflammatory and immune processes and responses (9). Microgravity alters the B/T cell ratio, and may cause an impaired immune response (10). To determine whether these variations in other systems contribute to losses in the musculoskeletal system during disuse, we employed NOD2 deficient mice. While contributions of single genes are pursued by employing genetic KO, the effects of small variations of the genome on bone's response to disuse are yet to be explored. WT-B6J and C57BL/6N (WT-B6N) are sub-strains of B6 mice that exhibit slight variations in their genome including 34 single nucleotide polymorphisms (SNP) and 2 insertion-deletion (INDELS) in coding regions and 15 gene linked chromosomal structural variations (SV).(11) We recently identified one SNP and one SV as potential modulators of bone's sensitivity to unloading. We therefore hypothesized that B6J and B6N will respond differently to hindlimb unloading (HLU). Methods and Results To determine the role of FGF2 and ESR1, 36 FGF2-/- (STOCK Fgf2tm2Doe/J), 36 ESR1-/- (B6.129P2-Esr1tm1Ksk/J), 20 NOD2-/- mice (B6.129S1-Nod2tm1Flv/J) null mice were divided into baseline controls, sacrificed at 9 weeks of age to quantify baseline phenotype (BC, n=4), age matched controls, sacrificed at 11 weeks to determine ambulatory cage control phenotype (AC, n=16 for ESR1 and FGF2 and n=8 for NOD-2) and hindlimb unloaded groups, sacrificed at 11 weeks after 14 days of hindlimb unloading (HLU, n=16 for ESR1 and FGF2 and n=8 for NOD-2). These mice were compared to wild type C57BL/6J (WT-B6J, n=34) that were divided into the same three groups described above (AC, HLU, n=12 and BC, n=10). Thirty-four, 9 week old female B6N mice were divided into groups similar to B6J mice. Bone from the distal metaphysis was analyzed via microCT scanning (Scanco µCT 40) at 10µm resolution. Bone indices within the same genotype were tested for statistical significance using one way ANOVA and Tukey post-hoc tests. Differences in baseline bone morphology (KO vs WT) were tested for significance using two-tailed, independent samples t-tests. Gene x Treatment (hindlimb unloading) interactions were analyzed using two-way ANOVA. At baseline, FGF2-/- mice had a 14% lower trabecular number (Tb.N) and 17% greater Tb.Th when compared to wildtype B6J mice (P<0.05). All other trabecular architectural indices remained similar. However, these micro-architectural differences were abolished as the mice aged (at 11 weeks). Following 2 weeks of hindlimb unloading, the response of the trabecular compartment of the distal metaphysis was similar between FGF2 null mice and WT-B6J mice. At baseline, ESR1-/- mice had a 43% greater trabecular bone volume fraction (BV/TV), 32% greater trabecular number (Tb.N), and trabecular spacing (Tb.Sp) was 25% lower when compared to wildtype B6J mice (P<0.05). Similar differences were seen in all other trabecular architectural indices except trabecular thickness (Tb.Th) which did not change. These differences in micro-architectural differences persisted as the mice aged (at 11 weeks). Following 2 weeks of hindlimb unloading, the response of the trabecular compartment of the distal metaphysis was severely stunted in ESR1 null mice with just a 8% loss in Tb.Th, while the WT-B6J mice had 19% lower thickness compared to age matched controls (Gene x Treatment, P<0.01). Similar trends were seen in losses of BV/TV, and Tb.N between ESR1-/- and WT-B6J mice (12 % vs 32% and 2% vs 5%). However these differences did not reach statistical significance. At baseline, NOD2-/- mice had 35% lower trabecular BV/TV, 14% lower Tb.N, and Tb. Sp was 17% greater (all P<0.01). Similar trends persisted as mice aged. Following hindlimb unloading, the degree and pattern of loss in trabeculae of the distal femoral compartment was similar between NOD2-/- mice and WT-B6J mice. In the metaphysis, at baseline, B6N mice had 21% lower bone volume (BV), bone volume fraction and 6% lower trabecular number and thickness when compared to B6J mice (all P<0.05) while tissue volume (TV) and connectivity density (Conn.D) were similar between strains. Upon HLU, tissue losses did not significantly differ between B6J and

B6N in the trabecular metaphysis when compared to respective age matched control mice.

Discussion

Together, these data emphasize the importance of FGF2, ESR1, and NOD2 in maintaining trabecular architecture under physiological loading conditions. Results also suggest that ESR1 plays a large role in modulating the sensitivity of the trabecular compartment to the lack os mechanical signals). The contribution of FGF2 and NOD2 towards influencing bone's response to disuse was mostly insignificant. Finally, this data highlights the importance of subtle variations in the genome (B6J vs B6N) in regulating complex bone morphology and microarchitecture.

Currently, we are pursuing the next three target genes (ESR2, PAK6 and FAS) on our shortlist.

References

1. COFFIN J., FLORKIEWICZ R., NEUMANN J., MORT-HOPKINS T., DORN G. N., LIGHTFOOT P. et al. Abnormal bone growth and selective translational regulation in basic fibroblast growth factor (FGF-2) transgenic mice, Molecular biology of the cell 1995: 6: 1861-1873.

2. MONTERO A., OKADA Y., TOMITA M., ITO M., TSURUKAMI H., NAKAMURA T. et al. Disruption of the fibroblast growth factor-2 gene results in decreased bone mass and bone formation, Journal of Clinical Investigation 2000: 105: 1085-1093.

3. XIAO L., LIU P., LI X., DOETSCHMAN T., COFFIN J. D., DRISSI H. et al. Exported 18-kDa isoform of fibroblast growth factor-2 is a critical determinant of bone mass in mice, Journal of Biological Chemistry 2009: 284: 3170-3182.

4. LIANG H., PUN S., WRONSKI T. Bone Anabolic Effects of Basic Fibroblast Growth Factor in Ovariectomized Rats 1, Endocrinology 1999: 140: 5780-5788.

5. HURLEY M., OKADA Y., XIAO L., TANAKA Y., ITO M., OKIMOTO N. et al. Impaired bone anabolic response to parathyroid hormone in Fgf2-/- and Fgf2+/- mice, Biochemical and biophysical research communications 2006: 341: 989-994.

6. MARCUS R. Post-menopausal osteoporosis, Best practice & research Clinical obstetrics & gynaecology 2002: 16: 309-327.

7. LANYON L., ARMSTRONG V., ONG D., ZAMAN G., PRICE J. Is estrogen receptor alpha key to controlling bones' resistance to fracture?, Journal of Endocrinology 2004: 182: 183-191.

8. LEE K., JESSOP H., SUSWILLO R., ZAMAN G., LANYON L. The adaptive response of bone to mechanical loading in female transgenic mice is deficient in the absence of oestrogen receptor-alpha and-beta, Journal of endocrinology 2004: 182: 193-201.

9. YUAN H., ZELKA S., BURKATOVSKAYA M., GUPTE R., LEEMAN S. E., AMAR S. Pivotal role of NOD2 in inflammatory processes affecting atherosclerosis and periodontal bone loss, Proceedings of the National Academy of Sciences 2013: 110: E5059-E5068.

10. GAIGNIER F., SCHENTEN V., BITTENCOURT M. D. C., GAUQUELIN-KOCH G., FRIPPIAT J.-P., LEGRAND-FROSSI C. Three Weeks of Murine Hindlimb Unloading Induces Shifts from B to T and from Th to Tc Splenic Lymphocytes in Absence of Stress and Differentially Reduces Cell-Specific Mitogenic Responses, PloS one 2014: 9: e92664.

11. SIMON M. M., GREENAWAY S., WHITE J. K., FUCHS H., GAILUS-DURNER V., WELLS S. et al. A comparative phenotypic and genomic analysis of C57BL/6J and C57BL/6N mouse strains, Genome biology 2013: 14: R82.

Bibliography Type:	Description: (Last Updated: 01/16/2018)
Abstracts for Journals and Proceedings	Sankaran J, Lublinsky S, Zhang W, Donahue LR, Judex S. "Genetic regulation of bone morphology is strongly non-uniform within a single bone." 35th Annual Meeting of the American Society for Bone and Mineral Research, Baltimore, Maryland, October 4-7, 2013. J Bone Miner Res 2013;28(Suppl 1). , Oct-2013
Abstracts for Journals and Proceedings	Sankaran J, Zhang W, Donahue LR, Judex S. "CD44-/- and ESR1-/- mice exhibit contrasting patterns of bone loss during hindlimb unloading." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3269.pdf , Feb-2014
Abstracts for Journals and Proceedings	Sankaran J, Tuthill A, Donahue LR, Judex S. "Differences in bone microarchitecture and the degree of bone loss during hindlimb unloading between C57Bl/6J and C57Bl/6N mice." 36th Annual Meeting of the American Society for Bone and Mineral Research, Houston, Texas, September 12-15, 2014. J Bone Miner Res 29(Suppl 1). , Sep-2014
Abstracts for Journals and Proceedings	Sankaran J, Kuriakose S, Zhang,W, Donahue LR, Judex S. "Differential patterns of bone loss in the femur and vertebra of CD44-/- mice during hindlimb unloading." 36th Annual Meeting of the American Society for Bone and Mineral Research, Houston, Texas, September 12-15, 2014. J Bone Miner Res 29(Suppl 1). , Sep-2014
Articles in Peer-reviewed Journals	Ozcivici E, Zhang W, Donahue LR, Judex S. "Quantitative trait loci that modulate trabecular bone's risk of failure during unloading and reloading." Bone. 2014 Jul;64:25-32. Epub 2014 Mar 31. http://dx.doi.org/10.1016/j.bone.2014.03.042 ; PubMed PMID: 24698783 , Jul-2014

Articles in Peer-reviewed Journals	Ozcivici E, Judex S. "Trabecular bone recovers from mechanical unloading primarily by restoring its mechanical function rather than its morphology." Bone. 2014 Oct;67:122-9. http://dx.doi.org/10.1016/j.bone.2014.05.009 ; PubMed PMID: 24857858 , Oct-2014